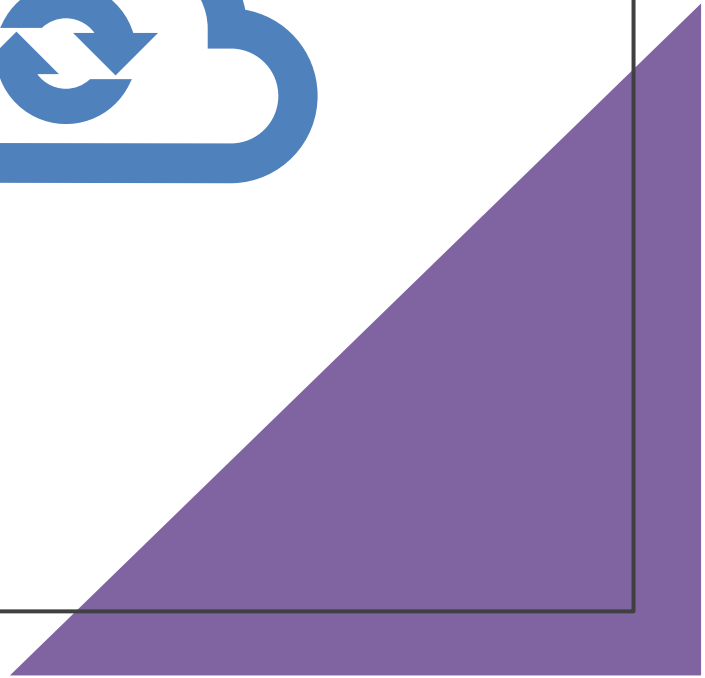
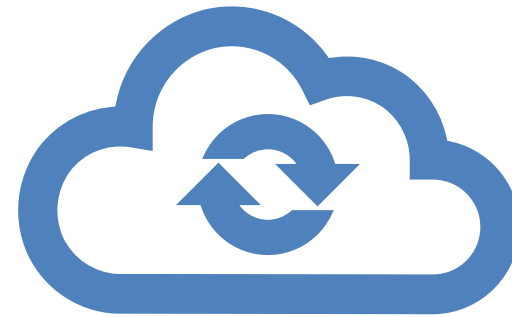


Optimizing CI/CD for Hybrid and Cloud Environments with DevOps and AI

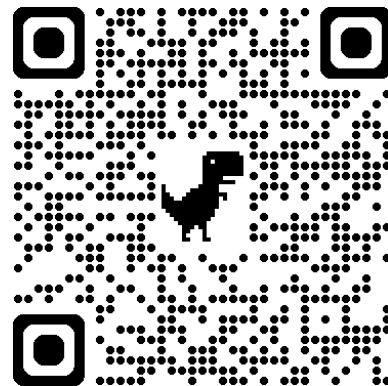
Building scalable, secure, and AI-enhanced CI/CD pipelines



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Agenda



FOUNDATIONS OF MODERN CI/CD

LEVERAGING AI IN CI/CD PIPELINES

SECURITY AND GOVERNANCE

KEY PRACTICES AND PATTERNS

DEMO AND WALKTHROUGH

Q&A AND TAKEAWAYS

Foundations of Modern CI/CD

Principles: Continuous Integration and Delivery



Key tools for SCM, CI/CD, Testing, and Observability

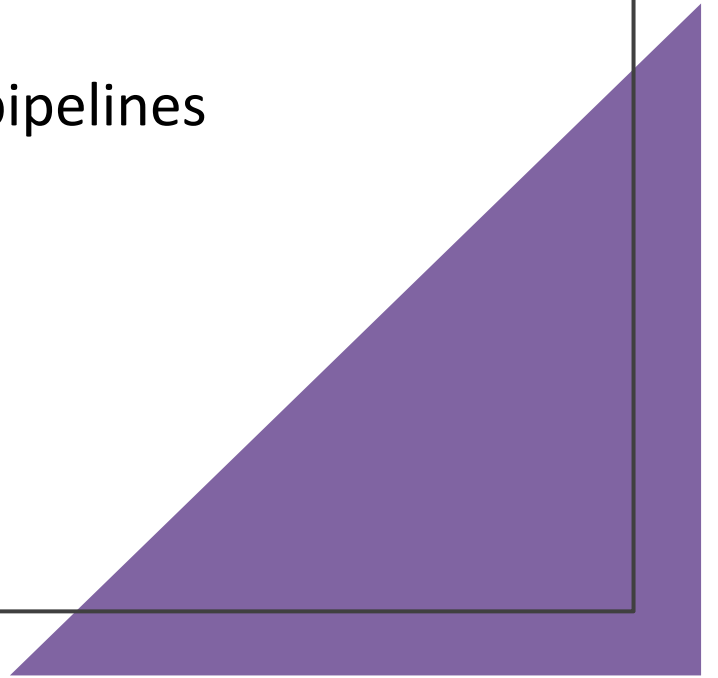


Automation as the backbone of modern pipelines



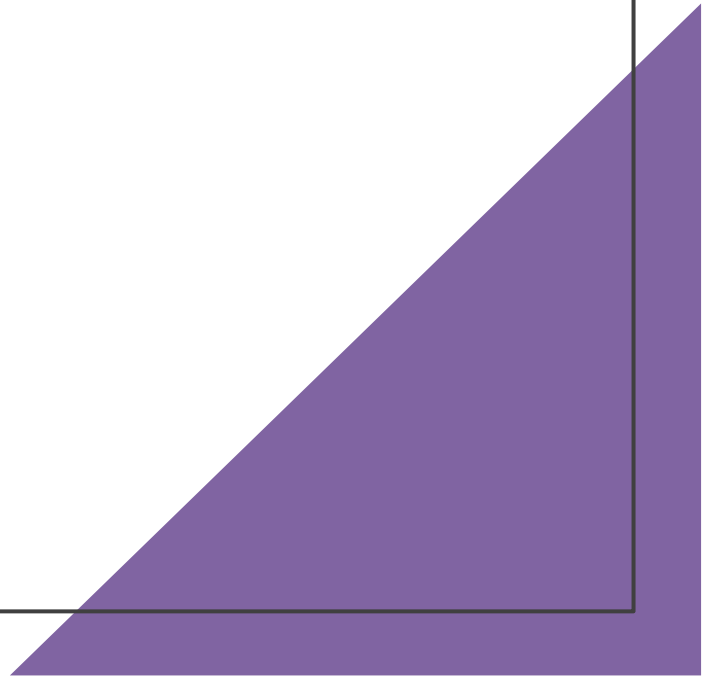
Leveraging AI in CI/CD Pipelines

- Predictive build and test optimization
- AI-assisted code review and recommendations
- Anomaly detection in pipelines and deployments
- Smarter rollbacks and deployment decisions



Security and Governance

- Mitigating AI-related vulnerabilities
- Ensuring model integrity and compliance
- Zero Trust pipelines
- Policy-as-Code and automated compliance checks

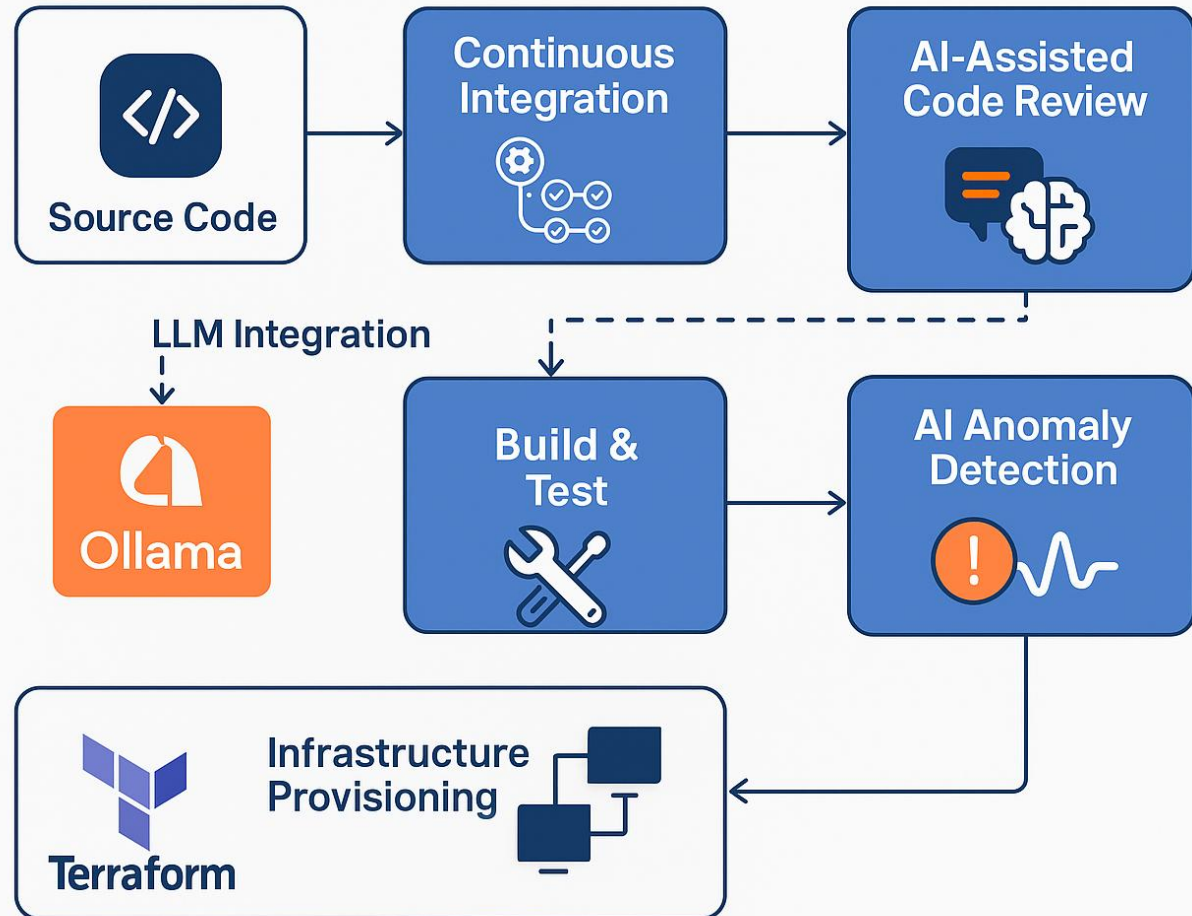


Key Practices and Patterns

- GitOps for declarative pipelines
- Infrastructure as Code with Terraform, Pulumi, or Bicep
- Progressive delivery: Canary and blue-green deployments
- Feature flag strategies



Example Workflow



Demo Walkthrough



Push → AI-assisted code review
→ Automated testing



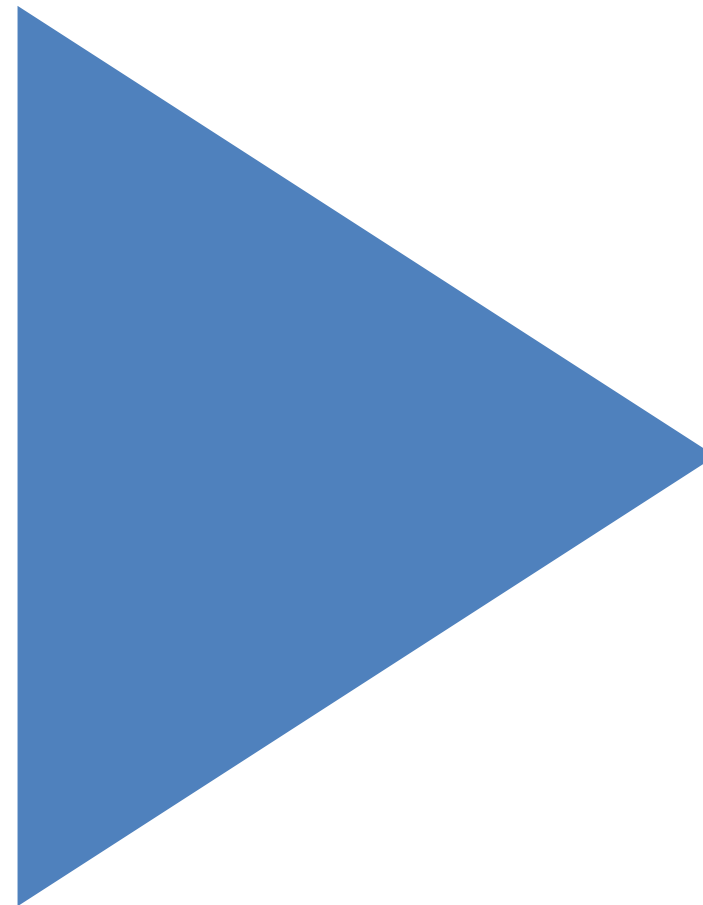
AI anomaly detection → Canary
deployment



Tools: GitHub Actions,
Terraform, ArgoCD, LLM
integration



DEMO



Scripts and Demo Sheet Available at:
<http://github.com/Dbakevlar/Presentations2025/>

Key Takeaways



Automate everything, with governance baked in



Balance AI innovation with security and compliance



Use observability for smarter, safer decisions



Start small, scale responsibly



Q&A

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